Exact Profile Robustness for Hellinger Contraction with Arbitrary Boolean Output Bias

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M238

October 8, 2026

Abstract

We examine the finite scalar certificate in the September 24 manuscript *Hellinger contraction with arbitrary Boolean output bias*, separating its profile module from its pair inequalities, dimension induction and semigroup continuation. An exact odd perturbation preserves the reflection constraint and all three normalization values of each of its 17 profiles. We derive the coupled first and second variations of the closure polynomial, and certify whole parameter intervals by minimizing each quadratic Bernstein coefficient, retaining the positive-sign guard required for noise-endpoint reduction. An independently implemented rational calculation certifies row-specific tilt radii and the common radius 2^{-15}; this permits changing each first integer Chebyshev coefficient by any integer between -3051 and 3051. The original profile/base-case program is separately replayed. A general residual budget explains what survives under other normalized perturbations and why absolute value or endpoint shortcuts can fail. The result is a robust finite profile family, not an independent certification of the global Hellinger or most-informative Boolean function claim. Contribution assessment: C3/5, a source-specific exact family of robust profile certificates with coupled closure and base-case margins, retaining the independent pair and continuation premises.

Contents

1 Claim, conventions and the actual proof boundary **2**
1.1 Current public context . . . 2
1.2 Whole-source dependency map . . . 3

2 The scalar-to-induction interface **3**
2.1 What the pair certificate supplies . . . 4
2.2 The five profile tests . . . 5

3 Signed margins and exact finite enclosures **6**
3.1 Bernstein enclosure, with no sampled-domain inference . . . 6
3.2 A retained square-root sign guard . . . 6

4 Coupled profile perturbations **7**

5 An explicit robust family for all 17 source rows **8**
5.1 Base cases and why normalization is essential . . . 8
5.2 Exact finite calculation and its tolerance table . . . 8

6 A residual budget beyond the single odd direction **10**
6.1 A degree-aware coefficient route 11

7 Why the local certificate is not the global theorem **11**
7.1 Three retained continuation mechanisms 12

8 Evidence, retained mechanisms and limitations **12**
8.1 What was actually executed 12
8.2 Reading and provenance boundary 13
8.3 What is retained, and what is corrected 13

9 Advantages, costs and feasible next steps **13**

1 Claim, conventions and the actual proof boundary

Let X be uniform on $\{-1,1\}^n$, $f:\{-1,1\}^n\to\{-1,1\}$, $m=\mathbb{E}f$, and T_ρ the Boolean noise operator with independent coordinate correlation ρ. Write

$$H(t)=\sqrt{1-t^2},\qquad s=\sqrt{1-\rho^2}.$$

The assigned source [1] claims, for every $n\geq 1$ and $\rho\in[-1,1]$,

$$H(m)-\mathbb{E}H(T_\rho f)\leq 1-s. \tag{1}$$

Here "arbitrary bias" is the bias of the Boolean *output*; the input measure remains uniform. Constants and signed coordinates must be retained as exact cases. Symmetry reduces to $0\leq\rho\leq 1$, and limits cover the noise endpoints. For a nonconstant Boolean function and $0<\rho<1$, the strictly positive noise kernel gives $|T_\rho f|<1$.

The source's low-noise induction uses $0<s\leq 21/25$, seven fixed pair parameter groups and 17 polynomial profiles. Continuation treats $21/25\leq s<1$ by three bias regimes. This report freshly read every cached scientific source file, but does not infer that every imported analytic estimate or executable certificate has thereby been independently verified. Its proved increment concerns only replacement of the 17 profiles by a quantified family. In particular, no unrestricted Bellman inequality for arbitrary couplings is inferred from these profile computations.

1.1 Current public context

Chen, Gohari and Nair's August 5, 2026 revision [2] states the Hellinger conjecture, derives its semigroup derivative in Lemma 6 and gives the conditional differential criterion in Theorem 7. Their proposed ordinary-information route still has explicitly conjectural inequalities; the natural unrestricted Hellinger candidate also has a counterexample. Their time is $\rho=e^{-2t}$, whereas ours below is $\rho=e^{-\tau}$. The scope matters: a finite-dimensional reduction is not itself its verification. We inspected these statements and the displayed derivative proof, not all of that paper's proofs.

The September 21, 2026 preprint of Chen and coauthors [3] claims a computer-assisted proof of the ordinary Courtade–Kumar conjecture; its abstract and metadata were inspected, not its long proof or supplement. Durcik, Ivanisvili, Roos and Xie's February 2026 preprint [4] reports the critical-exponent cube inequality and the balanced low-noise Hellinger limit. Its abstract was inspected; that statement is not the all-noise, all-output-bias inequality (1). The assigned September 24 claim and these recent claims are attributed as such, not declared independently accepted or certified by this report.

1.2 Whole-source dependency map

The map below distinguishes literal reading from the narrower independent calculation. All listed files are under the assigned source's `build/sections/` directory.

Source module	Mechanism and retained obligation
`preliminaries`	Uniform measure, Fourier normalization, noise and edge integration by parts.
`induction`	Compression, least-positive coordinate selection, small active-coordinate bases, root monotonicity, dimension induction.
`scalar`	Three endpoint/interior pair charts, divided-difference arithmetic, 17 profiles, positive-sign and base-case certificates.
`continuation`	First-crossing formulation along the noise semigroup and reduction to three bias regimes.
`central`	Calibrated two-point comparison, variance/energy resources, first-level spreading and the dictator exceptional case.
`intermediate`	Parity gain, tangential profile and pointwise two-variable certificates with positive radical branches.
`large-bias`	Forward/reverse norm estimates, Boolean moment control, finite series plus an explicitly bounded remaining tail.
`arithmetic`	Exact integer/scaled-interval rules and the finite certificate interfaces; rejection or subdivision on failed guards.
`conclusion`	Assembly and one-way implication to ordinary mutual information in bits.

The main file, figures, typography definitions, references, metadata and profile tables were also read. The original private volume16 clearance chapter [5] was freshly read in full: its power-to-Bernstein identity, convex-hull proof, subdivision and separation of model error from numerical enclosure are the relevant original comparison, not a later report. Neither its physical clearance setting nor a small sampled residual automatically supplies a cube inequality.

2 The scalar-to-induction interface

For an increasing Boolean function with at least four active coordinates, the source selects a coordinate with positive correlation x satisfying

$$0 < x \le b(m) := \min\{3/8,\ 9/16 - |m|/2,\ 1 - |m|\}. \tag{2}$$

The two section means are $m - x, m + x$. We include $x = 0$ by continuity in all polynomial certificates and set $\mathcal{D} = \{(m, x) : -1 \le m \le 1,\ 0 \le x \le b(m)\}$. In particular $m \pm x \in [-1, 1]$.

For one entire noise row $s \in [s_0, s_1]$, fix $L, U, V, \alpha, \Lambda, \mu$ and let

$$\begin{aligned} \lambda(h) &= L + Uh + Vh^2, \\ G(p) &= H(2p-1)\left(1 + \frac{2p-1}{1 + H(2p-1)\lambda(H(2p-1))}\right). \end{aligned} \tag{3}$$

The seven source parameter groups are given below; all displayed entries are integers divided by 1000. The left endpoint of each group is the previous right endpoint, starting at zero.

$1000s_{\text{right}}$	$1000L$	$1000U$	$1000V$	1000α	1000Λ	1000μ
70	940	-440	300	960	1000	956
160	940	-440	300	930	1050	953
300	886	-398	250	860	1150	985
490	842	-335	180	770	1150	1024
630	842	-335	180	690	1150	1058
750	842	-335	180	632	1160	1087
840	842	-335	180	583	1160	1111

Since $\lambda \geq L + U \geq .488$, all denominators in (3) are positive. One retains $G \geq 0$, $G(0) = G(1) = 0$, and $G(p) + G(1-p) = 2H(2p-1)$.

2.1 What the pair certificate supplies

For $0 \leq p_0 \leq p_1 \leq 1$, let $z = p_1 - p_0$, $\delta = (1-\rho)/2$, and $p_t = p_0 + tz$. Define

$$D = \frac{G(p_\delta) + G(p_{1-\delta}) - G(p_0) - G(p_1)}{2}, \qquad S = \alpha G(p_1) + (1-\alpha)G(p_0).$$

The source's independent pair obligations are

$$D \geq 0, \qquad D(S + \Lambda D) \geq \mu s^2 \left(z - \frac{G(p_1) - G(p_0)}{4L} \right)^2. \tag{4}$$

Its scalar proof does not simply sample this expression. Endpoint coordinates factor the vanishing terms before interval evaluation, and divided differences extend to coincident nodes. For example, for tuples at nodes $0, 1, 2$, the exact identities

$$(ab)_{012} = a_0 b_{012} + a_{01} b_{12} + a_{012} b_2, \qquad (\sqrt{c})_{012} = \frac{c_{012} - a_{01} a_{12}}{a_0 + a_2}$$

hold when the root denominators are positive. Invalid reciprocal/root guards trigger subdivision, never acceptance. The source uses 21 initial boxes; one interior case permits depth 14. That C++ computation is *not replayed here*. Concavity of G also comes from the certified endpoint charts, not from an assumption silently added to this report.

After averaging (4) over the remaining cube and applying Cauchy–Schwarz with its nonnegative factors, write

$$y = \mathbb{E}G(p_1)/s, \quad t = \mathbb{E}G(p_0)/s, \quad e = \mathbb{E}D/s, \quad k = \frac{s}{2L}, \quad Q = \alpha y + (1-\alpha)t.$$

Then

$$e(Q + \Lambda e) \geq \mu\{x - k(y-t)/2\}^2. \tag{5}$$

The source's root map

$$\mathcal{R}_x(y,t) = \frac{y+t}{2} + \frac{\sqrt{Q^2 + 4\Lambda\mu\{x - k(y-t)/2\}^2} - Q}{2\Lambda} \tag{6}$$

is coordinatewise nondecreasing. A sufficient retained bound is $1 - \max(\alpha, 1-\alpha)/\Lambda - k\sqrt{\mu/\Lambda} > 0$; the source verifies it for these seven groups. This property is not affected by changing only the polynomial profile.

2.2 The five profile tests

The source writes, with exact integer c_{ij} and $D_* = 10^8$,

$$P_i(m) = 1 + D_*^{-1} \sum_j c_{ij}\{T_j(m) - T_j(0)\}, \qquad B_i(m) = (1-m^2)P_i(m). \tag{7}$$

Thus $B_i(0) = 1$ and $B_i(\pm 1) = 0$. For a fixed profile suppress i and define the polynomial quotients

$$\begin{aligned}
E_B(m) &= \frac{(B(m) + B(-m))/2 - 1}{m^2}, \\
C(m,x) &= \frac{B(m+x) - B(m-x)}{2x}, \\
A(m,x) &= \frac{2B(m) - B(m+x) - B(m-x)}{2x^2}, \\
J_s(m,x) &= 1 - \frac{sC(m,x)}{2L}, \\
R(m,x) &= B(m) + (2\alpha - 1)xC(m,x) + (\Lambda - 1)x^2 A(m,x), \\
V_s(m,x) &= \mu J_s(m,x)^2 - A(m,x)R(m,x).
\end{aligned} \tag{8}$$

$$\tag{9}$$

No division is performed at zero: the numerators factor in the polynomial ring, giving $C(m,0) = B'(m)$, $A(m,0) = -B''(m)/2$. The scalar module certifies

$$P > 0, \quad 1 + s_1 E_B(m)(1 + H(m)) > 0, \quad J_{s_1} > 0, \quad V_{s_0} > 0, \quad V_{s_1} > 0, \tag{10}$$

on $[-1,1]$, $[0,1]$ and $\mathcal{D}$, respectively. Its small-cube base inequalities are an additional obligation, not a sixth polynomial sign that can be omitted.

Lemma 2.1 (Endpoint reduction with its necessary guard)**.** *Fix one complete row, including its profile and all six parameters. If $J_{s_1} > 0$ on $\mathcal{D}$, then $J_s > 0$ for $0 \le s \le s_1$. For fixed (m,x), the minimum of V_s on $[s_0, s_1]$ is attained at an endpoint. Moreover the second inequality in (10) implies the reflection condition*

$$\frac{s}{2}\{B(m) + B(-m)\} \ge H(m) - 1 + s \quad (0 \le s \le s_1). \tag{11}$$

Proof. J_s is affine in s, and $J_0 = 1$. While $J_s > 0$, the derivative of μJ_s^2 is $-\mu C J_s/L$, with constant sign because C is fixed. The subtracted term AR is independent of s. This proves the endpoint claim; it is not a theorem about arbitrary quadratic polynomials. For reflection subtract the right side to get

$$m^2 \left(sE_B(m) + \frac{1}{1 + H(m)} \right).$$

The bracket is even in m. If $E_B \ge 0$ it is positive; otherwise its minimum is at $s = s_1$. At $m = 0$ the original equality is exact. □

Proposition 2.2 (Closure from the signs)**.** *Assume $B \ge 0$ on $[-1,1]$, the pair/root premises above, and (10). Then $\mathcal{R}_x(B(m+x), B(m-x)) \ge B(m)$ throughout $\mathcal{D}$. Consequently replacement profiles satisfying these same obligations and the source's small-cube bases can be used in the source dimension induction.*

Proof. Let $y = B(m+x), t = B(m-x)$. Their average is $B(m) - x^2 A$, and $Q = B(m) - x^2 A + (2\alpha - 1)xC \geq 0$. If $A \leq 0$, the average already dominates $B(m)$ and the added root is nonnegative. If $A > 0$, the nonnegative root e of $e(Q + \Lambda e) = \mu x^2 J_s^2$ dominates $x^2 A$ precisely when

$$\mu x^2 J_s^2 \geq x^2 A(Q + \Lambda x^2 A) = x^2 AR.$$

For $x > 0$ this is $V_s \geq 0$; $x = 0$ follows by continuity. Root monotonicity permits insertion of the section lower bounds from dimension $n-1$. The bases start the induction, and reflection supplies the stated lower bound for the average of the two signs. This is a conditional interface proof: it does not prove (4). □

3 Signed margins and exact finite enclosures

3.1 Bernstein enclosure, with no sampled-domain inference

For a polynomial $q(u) = \sum_{j=0}^{N} a_j u^j$ on $[0,1]$, the identity

$$q(u) = \sum_{k=0}^{N} \beta_k \binom{N}{k} u^k (1-u)^{N-k}, \qquad \beta_k = \sum_{j=0}^{k} a_j \frac{\binom{k}{j}}{\binom{N}{j}} \tag{12}$$

follows by expanding $u^j = u^j(u + (1-u))^{N-j}$ and comparing coefficients. The basis is nonnegative and sums to one; tensor products give the same convex-hull conclusion on rectangles. Restriction by de Casteljau subdivision is exact. These are classical facts, also proved in the freshly read private original [5]; they are not the contribution claimed for M238.

The upper boundary of $\mathcal{D}$ has knots

$$(-16, 0),\ (-14, 2),\ (-6, 6),\ (6, 6),\ (14, 2),\ (16, 0),$$

with both coordinates divided by 16. A consecutive pair $(a, c), (b, d)$ is covered exactly by

$$m = \frac{a + (b-a)u}{16}, \qquad x = v\frac{c + (d-c)u}{16}, \qquad (u, v) \in [0,1]^2. \tag{13}$$

If a polynomial in (m, x) has total degree at most N and x degree at most K, the transformed bidegree is at most (N, K). The order N must be chosen *before* treating this as a rectangle certificate; using only its original m degree can miss terms from x^j. All lower coefficients below use exact rational arithmetic, not decimal evaluation at a mesh.

3.2 A retained square-root sign guard

Suppose e is a lower Bernstein bound for E_B on $[a, b] \subset [0,1]$. If $e \geq 0$, reflection is automatic. If $e < 0$, write $q = 1 + s_1 e$ and $z = -s_1 e\sqrt{1-a^2} \geq 0$. The sufficient tests are

$$q > 0, \qquad \Delta = q^2 - (s_1 e)^2(1 - a^2) > 0. \tag{14}$$

Indeed $E_B(m) \geq e$ and $H(m) \leq H(a)$ imply

$$1 + s_1 E_B(m)(1 + H(m)) \geq q - z = \frac{\Delta}{q + z} \geq \frac{\Delta}{2q} > 0. \tag{15}$$

This rational lower margin permits a further downward perturbation of E_B by less than $\Delta/(4s_1 q)$ on this panel when $s_1 > 0$. It is merely a sufficient local budget; all panels and positive-e branches still have to be covered.

For example $s_1 = 1, e = -4, a = 3/4$ gives $q = -3$ and $q^2 - z^2 = 2 > 0$, but the required expression is $-3 - \sqrt{7} < 0$. Squaring without the first guard is false. Similarly $(s - 1/2)^2 - 1/10$ is positive at $s = 0, 1$ but negative at $1/2$. Lemma 2.1, not an arbitrary quadratic-endpoint rule, justifies the source's two noise-endpoint tests.

4 Coupled profile perturbations

Let h be a real polynomial satisfying $h(0) = h(1) = h(-1) = 0$ and set $B_\theta = B + \theta h$. When needed write $h = (1 - m^2)p_h$ with $p_h(0) = 0$, so $P_\theta = P + \theta p_h$. Use the linear operators in (8) to define A_h, C_h, E_h, and put

$$R_h = h + (2\alpha - 1)xC_h + (\Lambda - 1)x^2 A_h, \qquad j_h = -\frac{sC_h}{2L}.$$

Proposition 4.1 (Exact dependent variation)**.** *For fixed row parameters, the perturbed closure polynomial is*

$$\begin{aligned} V_{s,\theta} &= V_s + \theta\mathcal{L}_s(h) + \theta^2\mathcal{Q}_s(h), \\ \mathcal{L}_s(h) &= -\frac{\mu s}{L}J_s C_h - AR_h - A_h R, \\ \mathcal{Q}_s(h) &= \frac{\mu s^2}{4L^2}C_h^2 - A_h R_h. \end{aligned} \tag{16}$$

In particular the errors in A, C, J, R, V are not independently adjustable.

Proof. Linearity gives $A_\theta = A + \theta A_h$, $C_\theta = C + \theta C_h$, $R_\theta = R + \theta R_h$ and $J_{s,\theta} = J_s + \theta j_h$. Expand $\mu(J_s + \theta j_h)^2 - (A + \theta A_h)(R + \theta R_h)$. There are no omitted higher terms. □

Theorem 4.2 (Quadratic coefficient certificate)**.** *Fix $t > 0$. For each endpoint $s = s_0, s_1$ and each panel in a finite cover of $\mathcal{D}$, expand the three polynomials in (16) in a* common *tensor Bernstein basis of adequate degree. Denote one triple of coefficients by (v, ℓ, q) and define*

$$\mathfrak{m}_t(v, \ell, q) = \begin{cases} \min\{v - |\ell|t + qt^2,\ v - \ell^2/(4q)\}, & q > 0,\ |\ell| \le 2qt, \\ v - |\ell|t + qt^2, & \textit{otherwise.} \end{cases} \tag{17}$$

If every such minimum is positive, then $V_{s,\theta} > 0$ at both endpoints for all $|\theta| \le t$. If also the affine coefficient tests for P_θ and $J_{s_1,\theta}$ are positive, the same inequalities hold for the full spatial domain and the full noise row. Reflection and the bases must be checked separately.

Proof. The minimum of $v + \ell\theta + q\theta^2$ on $[-t, t]$ occurs at the worse endpoint, or at its vertex $-\ell/(2q)$ if $q > 0$ and that vertex lies in the interval. This is (17), including equality in the vertex-location test. For every fixed θ, the polynomial value is a convex combination of its Bernstein coefficients; hence it is at least the smallest of the positive minima. Affine coefficients use $a - t|b| > 0$. Positive $J_{s_1,\theta}$ and $J_{0,\theta} = 1$ then allow Lemma 2.1 for this perturbed profile. Compactness is used only after complete panel coverage, not to turn samples into a proof. □

This criterion is exact for each coefficient quadratic, but the Bernstein lower envelope is a sufficient, potentially conservative condition for the polynomial itself. Nor are the dyadic radii below maximal admissible profile perturbations. Positive quadratic terms have not been discarded as absolute errors, and an interior vertex has not been overlooked.

5 An explicit robust family for all 17 source rows

Take the particular normalized odd direction

$$h(m) = m - m^3, \qquad p_h(m) = m. \tag{18}$$

Its exact quotient data are

$$E_h = 0, \qquad A_h = 3m, \qquad C_h = 1 - 3m^2 - x^2, \tag{19}$$

and

$$R_h = m - m^3 + (2\alpha - 1)x(1 - 3m^2 - x^2) + 3(\Lambda - 1)mx^2.$$

Thus the potentially delicate reflection square-root certificate is *identical* for B and B_θ. The pair function G, all its endpoint singularity cancellations, its concavity proof, and the root monotonicity constants are also unchanged.

5.1 Base cases and why normalization is essential

The source treats constants, dictators and three-coordinate majority separately. They depend only on $B(\pm 1) = 0$ or $B(0) = 1$, which are unchanged. The remaining increasing functions on at most three active coordinates reduce, by the source's classification and reflection, to the up-sets $\{7\}, \{7, 6\}, \{7, 6, 5\}$ in binary coordinate order. Their means and dual means are $\pm 3/4, \pm 1/2, \pm 1/4$. The base expression is $\mathbb{E}G((1 + T_\rho f)/2)/s - B(m)$, with a removable limit at $s = 0$. Since G is fixed, its change is exactly $-\theta h(m)$. Consequently

$$|h(1/4)| = 15/64, \quad |h(1/2)| = 3/8, \quad |h(3/4)| = 21/64, \qquad \text{base loss} \le 3|\theta|/8. \tag{20}$$

No bound on G derivatives is required for this transfer.

For completeness, the replayed base code encloses, on 128 noise panels per row, the expressions obtained from

$$\delta = \frac{s^2}{2(1 + \sqrt{1 - s^2})}, \quad I_i = \frac{1}{4} \sum_{j:\, f(j) \neq f(i)} \delta^{d(i,j)-1}(1 - \delta)^{2-d(i,j)}.$$

It sets $e_i = s^2 I_i$, $h_i = 2\sqrt{I_i(1 - e_i)}$ and $q_i = f(i)(1 - 2e_i)/(1 + sh_i\lambda(sh_i))$; the two normalized bases are $\frac{1}{8}\sum_i h_i(1 \pm q_i) - B(\pm m)$. The code uses exact outward intervals at scale 10^{12}, including positive reciprocal guards. This avoids dividing by s at zero. Our new transfer uses the replayed source lower base margin $b_i/10000$; it does not present that replay as independently written base arithmetic.

5.2 Exact finite calculation and its tolerance table

The author program `M238-profile-robustness.py` parses only the integer tables from the pinned public script, not its executable code. It independently constructs the polynomials, maps the five trapezoids, converts to Bernstein form and subdivides by exact integer de Casteljau. For each of P, J, V_{s_0}, V_{s_1} it uses the source's maximum depths, stopping a panel when the unperturbed coefficients are all positive. The same panel partition and common degree are then used for all three variation coefficients. It checks (17) exactly.

The source integers use $\ell = 1000L$, $S = 1000s$, $\widehat{B} = D_* B$, and similarly for A, C. The program's scales are

$$\begin{aligned}
\widehat{J} &= 2\ell D_* - S\widehat{C} = (2\ell D_*)J_s, \\
\widehat{R} &= 1000\widehat{B} + (2000\alpha - 1000)x\widehat{C} + (1000\Lambda - 1000)x^2\widehat{A}, \\
\widehat{V} &= (1000\mu)\widehat{J}^2 - (2\ell)^2\widehat{A}\widehat{R} = 1000(2\ell)^2 D_*^2 V_s.
\end{aligned}$$

Chart clearing multiplies coefficients by the positive number $d^N N!K!$ for chart denominator d; each simultaneous nonconstant-axis half subdivision adds the positive factor 2^{N+K}. These factors are removed in the reported rational margins. No sign can change under this clearing, and no plotted floating-point value is used for acceptance.

Theorem 5.1 (Certified 17-profile odd-tilt family)**.** *Let B_i be the exact 17 source profiles, and choose independent real θ_i satisfying $|\theta_i| \le 2^{-n_i}$ with n_i in Table 1. Keep the entire source row's six parameters fixed. Then $B_i + \theta_i(m - m^3)$ obeys all five profile tests and the source's small-cube base inequalities on that row. In particular this holds for every $|\theta_i| \le 2^{-15}$ simultaneously. Conditional on the unchanged pair, compression/selection, classification and root premises, these profiles can replace the source profiles in its low-noise induction. This theorem does not independently establish the continuation regime.*

Proof. Equations (19) preserve reflection and the three exact normalizations. The exact coefficient computation in Theorem 4.2, reproducible by the accompanying program, gives the table's strictly positive endpoint closure margins and, across all rows at their stated radii, the rational lower bounds

$$P_\theta > \frac{1}{4}, \quad J_{s_1,\theta} > \frac{1}{5}, \quad V_{s_0,\theta}, V_{s_1,\theta} > \frac{19}{10^6}, \quad \text{remaining base margin} > \frac{1}{2000}. \tag{21}$$

The last bound is checked exactly as $b_i/10000 - 3 \cdot 2^{-n_i}/8$. The E-test and the original base margins are supplied by the separately successful exact source-profile replay. All polynomial sign tests here are finite integer/rational tests over complete Bernstein covers. Lemma 2.1 supplies every s in the row and Proposition 2.2 supplies the asserted conditional interface. At a shared noise endpoint either neighboring complete row is permitted; one may not combine one row's profile with the other's parameters. □

$1000s_0$	$1000s_1$	n_i	b_i	$10^6 \underline{v}_i$	$\deg P_i$
0	5	12	7	156	18
5	17	12	7	83	16
17	39	12	8	376	14
39	70	12	10	136	12
70	108	11	7	305	10
108	160	14	14	76	10
160	220	7	78	6048	8
220	300	7	69	3958	8
300	400	6	146	2230	8
400	490	8	224	5065	8
490	570	7	233	8580	8
570	630	9	255	1672	8
630	700	7	236	1361	10
700	750	9	222	235	10
750	800	8	191	1082	11
800	827	11	177	109	12
827	840	15	176	19	14

Table 1: Radius 2^{-n_i}, source base integer b_i, and a rational lower bound $\underline{v}_i$ for both endpoint closure polynomials after tilt. The displayed integer margins are rounded *down* from exact rationals; the exact unrounded margins and all counts are in the owned certificate JSON. The radii are sufficient, not optimal.

Corollary 5.2 (Independent integer coefficient edits)**.** *In each source row, replace c_{i1} by $c_{i1} + k_i$, keeping all other coefficients unchanged. Every independent choice $k_i \in \mathbb{Z}$ with $|k_i| \le 3051$ satisfies Theorem 5.1.*

Proof. $T_1(m) - T_1(0) = m$, so this changes B_i by $(k_i/10^8)(m - m^3)$. Exactly $3051 \cdot 32768 = 99\,975\,168 < 10^8$, whereas $3052 \cdot 32768 = 100\,007\,936 > 10^8$. Thus the asserted integers lie within the common certified radius. This is 6103^{17} coefficient tables, not that many independently run tests or new global theorems. □

Example 5.3 (A completely specified edit)**.** In the last profile, whose first numerator is 140455073, replace it by 140458124. This is $k = 3051$ and hence $\theta = 3051/10^8$. Its exact E_B is unchanged, $B(0) = 1$, $B(\pm 1) = 0$, and the base loss is at most $9153/(8 \cdot 10^8)$. The original last-row base margin is 176/10000, so that part has ample positive slack. The more restrictive closure check is not guessed from that base slack: Table 1 certifies both closure endpoints with lower margin $19/10^6$ for the larger interval $|\theta| \le 1/32768$.

6 A residual budget beyond the single odd direction

The exact tilt exploits a special cancellation. The following weaker, norm-based result gives an explicit route for more general profile rounding, rather than assuming that every small coefficient error is harmless. All norms here are uniform on $[-1, 1]$ or $\mathcal{D}$, as appropriate; their values require actual analytic bounds or complete enclosures.

Proposition 6.1 (Coupled sufficient error budget)**.** *Fix one row and suppose certified lower margins are $p_* > 0$ for P, $j_* > 0$ for J_{s_1}, $v_* > 0$ for both V_{s_0}, V_{s_1}, and $\gamma_* > 0$ for $1 + s_1 E_B(m)(1 + H(m))$ on $[0, 1]$. Let h be a polynomial vanishing at $0, \pm 1$. Put*

$$\eta_0 = \left\| h/(1 - m^2) \right\|_\infty, \quad \beta_0 = \|h\|_\infty, \quad \beta_1 = \|h'\|_\infty, \quad \beta_2 = \tfrac{1}{2} \|h''\|_\infty.$$

Let A_*, R_*, J_* be upper bounds for $|A|, |R|, |J_s|$, the last at both endpoints. With $X = 3/8$, define

$$\rho_h = \beta_0 + |2\alpha - 1| X \beta_1 + |\Lambda - 1| X^2 \beta_2, \qquad j_h^* = \frac{s_1 \beta_1}{2L}. \tag{22}$$

It suffices for replacing B by $B + h$ that

$$\begin{aligned} &\eta_0 < p_*, \qquad j_h^* < j_*, \qquad 2 s_1 \beta_2 < \gamma_*, \\ &2\mu J_* j_h^* + A_* \rho_h + R_* \beta_2 + \beta_2 \rho_h < v_*. \end{aligned} \tag{23}$$

The small-cube bases additionally require their original positive margin to exceed

$$\max_{m \in \{\pm 1/4, \pm 1/2, \pm 3/4\}} |h(m)|.$$

If h is odd, omit the condition involving γ_* altogether.

Proof. The segment $m + tx$, $-1 \le t \le 1$, remains inside $[-1, 1]$ by (2). The fundamental theorem of calculus twice gives

$$C_h(m, x) = \frac{1}{2} \int_{-1}^{1} h'(m + tx)\, dt, \qquad A_h(m, x) = -\frac{1}{2} \int_{-1}^{1} (1 - |t|) h''(m + tx)\, dt.$$

These identities also hold by limits at $x = 0$. Thus $|C_h| \le \beta_1$, $|A_h| \le \beta_2$, $|R_h| \le \rho_h$ and $|J_{s,\mathrm{new}} - J_s| \le j_h^*$. The even part of h has value and derivative zero at the origin, so the same integral remainder gives $|E_h| \le \beta_2$. Reflection therefore loses at most $2s_1\beta_2$; an odd h loses zero.

Finally, expansion as in Proposition 4.1 yields

$$V_{s,\mathrm{new}} - V_s = 2\mu J_s \delta J + \mu(\delta J)^2 - AR_h - A_h R - A_h R_h.$$

For a lower bound the nonnegative term $\mu(\delta J)^2$ may be discarded. The remaining absolute losses are exactly the last line of (23). This avoids charging a spurious negative square loss. The P and J tests follow directly; Lemma 2.1 extends the endpoint result. Base loss is exactly $-h(m)$ because G did not change. Constants and mean-zero cases retain their normalization. □

This proposition does not supply numerical norms for free, and it does not cover simultaneous changes of $G, L, U, V, \alpha, \Lambda, \mu$. Those alter the pair, root and base modules and require additional budgets. It is usually less sharp than the coefficient calculation: large power-basis coefficients can cancel in $AR_h + A_h R$, while the norm bound forgets their correlation.

6.1 A degree-aware coefficient route

For a normalized perturbation $h = (1 - m^2)\sum_{j\ge1} \varepsilon_j (T_j(m) - T_j(0))$, finite in degree, one may bound derivatives in a basis-aware way. Classical Chebyshev derivative bounds on $[-1, 1]$ give

$$U_0 = 2\sum_j |\varepsilon_j|, \quad U_1 = \sum_j j^2 |\varepsilon_j|, \quad U_2 = \sum_j \frac{j^2(j^2-1)}{3}|\varepsilon_j|,$$

and hence

$$\eta_0 \le U_0, \quad \beta_0 \le U_0, \quad \beta_1 \le 2U_0 + U_1, \quad \beta_2 \le U_0 + 2U_1 + U_2/2. \tag{24}$$

The last two follow by differentiating $(1 - m^2)p_h$; the bounds for T_j' and T_j'' can also be obtained from Markov's polynomial derivative inequalities applied to T_j. These classical estimates show the degree penalty explicitly; they are not a new approximation theorem. They also explain why a uniform coefficientwise tolerance for arbitrarily high degrees cannot be inferred from the single-mode radius 2^{-15}. Restricting to odd j retains $E_h = 0$ exactly. For several odd directions, one can instead keep the full quadratic form in their parameters and certify its minimum on a parameter box; the univariate vertex formula alone no longer solves that problem.

7 Why the local certificate is not the global theorem

It is useful to display the semigroup normalization rather than hiding it in a generic analytic remainder. Set $D_i u = (u(x) - u(x^{(i)}))/2$, $N = \sum_i D_i$ and $T_{e^{-\tau}} = e^{-\tau N}$. Under uniform measure,

$$\mathbb{E}\, uNv = \sum_i \mathbb{E}\, D_i u D_i v = \sum_{S\subseteq[n]} |S|\widehat{u}(S)\widehat{v}(S).$$

With $d = T_{e^{-\tau}} f$, $b = \mathbb{E}H(d)$ and $r = e^{-2\tau}$, differentiation on the finite cube gives

$$\frac{db}{d\tau} = \sum_i \mathbb{E}\, D_i d\, D_i\left(\frac{d}{H(d)}\right), \qquad \frac{ds}{d\tau} = \frac{r}{s}. \tag{25}$$

Indeed $\partial_\tau d = -Nd$ and $H'(d) = -d/H(d)$, so integration by parts gives the first identity with no missing factor of two. The finite state space and $|d| < 1$ justify differentiation away from endpoints. For a signed coordinate $d = \rho x_i$, the energy is $\rho^2/s = r/s$, which checks the normalization exactly.

If $F'(u) = H(u)^{-3/2}$ and $g = F(d)$, Cauchy–Schwarz on an edge gives

$$(a-b)\left(\frac{a}{H(a)} - \frac{b}{H(b)}\right) = (a-b)\int_b^a H(u)^{-3}\,du \geq (F(a)-F(b))^2.$$

Thus the energy in (25) is at least $\sum_i \mathbb{E}(D_i g)^2 \geq \operatorname{Var} g$. This elementary bound alone is not the source's continuation theorem. At a first downward crossing of $b = H(m) - 1 + s$ it is necessary to rule out energy at most r/s; mean-dependent corrections and strictness are essential.

7.1 Three retained continuation mechanisms

For $s \geq .84$ the source has $r \leq .2944$ and, by reflection, $m \geq 0$. The regimes and the obligations that remain imported here are:

1. For $m \leq r^2/2$, the central argument calibrates against $\pm x_0$ with $H(x_0) = b$. It retains the difference between the calibrated profile and the actual mean, adds an edge remainder beyond $\operatorname{Var} g$, and allocates both variance and first-Fourier-level spreading resources. The exceptional degree-one equality case is resolved as a dictator, not dismissed by a non-strict inequality. Its polynomial enclosures and strict resource allocation were read, not independently replayed here.

2. For $r^2/2 \leq m \leq .74$, even/odd decomposition gives extra energy from the even part. A tangential profile uses $\mathbb{E}V = 1 - s$, pointwise two-variable bounds and first-level Boolean moment constraints. The radical clearing keeps the sign assumptions on the physical domain; positivity of a cleared polynomial on a larger rectangle does not authorize division by a negative denominator there. These are separate certificates, not consequences of Table 1.

3. For $m \geq .74$, forward and reverse norm estimates control positive binomial moment expansions. A finite series is accompanied by an explicit tail mass, rather than dropped. The source obtains ratios strictly below one on four noise intervals. Its functional-inequality proofs and series remainder have been literally read but are not new results of this report.

The profile perturbation leaves these arguments unchanged if their premises are valid. It neither replaces them nor provides an independent audit of all their arithmetic. Likewise, the source's concave/convex change of entropy variable gives a one-way implication from the Hellinger statement to the ordinary information bound; an independent claim about the latter cannot be used in reverse to certify the former.

8 Evidence, retained mechanisms and limitations

8.1 What was actually executed

The unmodified pinned public `profile_certificate.py` was fetched with the following commit and verified Git blob:

Commit	`adc7f1241b42e322a6451854ab7e4b4c146bf78a`
Git blob	`ce70e3e77f21d5a581fa0a385b6295c4908680fc`

Its 189 lines were inspected before execution. It exited zero with `pass`; no changes were made to the original. Its SHA-256 is

`70aa31a4dfeaff3bdb8813228f316375b3ed1d7b42f00c0c98ae48cb680267b4.`

That replay covers 17 source profiles and their small-cube bases, not the original pair or continuation programs. It is execution of an existing checker, not an independent author test count.

The new profile-family computation uses 452 leaf panels and 317489 coefficient triples across P, J, V_{s_0}, V_{s_1}, not that many Boolean functions or unrelated theorems. It stores exact rational margins, row depths and input/script hashes. A separate bounded check program compares direct divided differences and closure evaluations with their polynomial forms, verifies tensor Bernstein evaluation and both half restrictions, tests the exact quadratic minimum formula, and checks positive-sign guards and the integer/base budgets. It passed 1604 exact cases comprising 7633 assertions. These finite checks are supplemental to the displayed algebraic proofs. Counts and assertion units are reported in the frozen evidence records.

8.2 Reading and provenance boundary

The read ledger lists each literal display interval, before/after SHA-256 and unread complement. All cached source texts were read, including the full induction, scalar, central, intermediate and large-bias scientific proofs. The private reading is exactly the original 299-line volume16 clearance chapter; the rest of the private corpus was not read for this assignment. The additional pinned public checker has its own read/replay receipt. Other public verification scripts, referenced papers' full proofs and any large external corpus are not credited as read or executed.

The parent's captured October 8 source-tree identity is provenance, not a live-branch check or a proof. Owned pre/post hashes are matched against that receipt where it covers the file; additional local metadata is identified separately. Parent reviews and checks are attributed to the parent reviewer and never added to author test counts. All private proof text stayed local; online queries used public titles or URLs only.

8.3 What is retained, and what is corrected

No substantive mathematical correction to the assigned source is asserted here. In particular, the source already retains polynomial normalization, the positive guard before squaring, whole-row selection at shared endpoints, divided-difference cancellation, the positivity of J_s, exact de Casteljau signs, and independent base cases. These are essential valid details, not discoveries credited to this report. The new content is the explicit odd-tilt family, the dependent closure variation, its coefficientwise quadratic certification and its measured 17-row tolerance/base ledger. Classical Bernstein enclosure, derivative bounds and scalar quadratic optimization retain their classical credit.

9 Advantages, costs and feasible next steps

The family theorem converts a single published table into a certified neighborhood in a meaningful direction: it directly tolerates quantization or intentional changes in the first Chebyshev numerator. Oddness removes the entire reflection error, normalization preserves exact equality cases, and unchanged G avoids regenerating the costly pair chart certificates. The base-case charge is exact and elementary. The coupled variation avoids treating dependent profile quantities as independent numerical inputs.

There are corresponding costs. The test still needs complete spatial coverage and exact coefficient arithmetic; 317489 coefficient triples are not a complexity improvement over the original checker. The reported radii depend on this atlas and these 17 profiles. They need not be sharp, and failure at twice a dyadic radius may reflect the Bernstein enclosure rather than actual loss of the inequality. High degree perturbations can have large derivative costs despite small coefficient errors. Changing the pair function or seeking a better noise split requires new work. No runtime speedup, new channel extremizer, strengthened global contraction constant or independently verified Courtade–Kumar theorem is claimed.

Useful further work is sharply delineated. One can optimize exact coefficient quadratic roots instead of searching dyadic radii, allow asymmetric intervals for a chosen signed rounding rule, and refine only the limiting leaf panels. A finite collection of odd modes can be handled by rigorous quadratic-form bounds on a parameter polytope, retaining cross terms and the same small-cube budget. Even modes require a genuine positive reflection margin using (15). A separate effort could audit and replay the 21 pair boxes and all continuation checkers. None of those unperformed tasks is counted as a present contribution.

Applicability remains L3: exact finite scalar certificates have a direct but limited connection to the original analytic enclosure methods. The independent contribution score is C3 for the explicit source-specific robust profile family and its full coupled margin transfer, not for the classical enclosure identity or the global conjectural claim.

References

[1] OpenAI, *Hellinger contraction with arbitrary Boolean output bias*, September 24, 2026. Assigned manuscript and public verification package, pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Hellinger-contraction-with-arbitrary-Boolean-output-bias-September-24-2026.

[2] Z. Chen, A. Gohari and C. Nair, *A Differential Equation Approach to the Most-Informative Boolean Function Conjecture*, arXiv:2502.10019v2, August 5, 2026. https://arxiv.org/abs/2502.10019v2.

[3] Z. Chen, A. Gohari, A. Javanmard, H. Lin, V. Mirrokni, C. Nair and D. P. Woodruff, *A Proof of the Most Informative Boolean Function Conjecture*, arXiv:2609.24931v1, September 21, 2026. Abstract/metadata context only. https://arxiv.org/abs/2609.24931v1.

[4] P. Durcik, P. Ivanisvili, J. Roos and X. Xie, *Sharp isoperimetric inequalities on the Hamming cube II: The critical exponent*, arXiv:2602.20462v1, February 24, 2026. Abstract/metadata context only. https://arxiv.org/abs/2602.20462v1.

[5] M. Subbarao, original private RamCalBooks, volume16, `chapters/ch05d-clearance.tex`, lines 1–299. Local confidential original; Bernstein hull/subdivision proof, model-error limits and margin accounting. No public upload.